



Multiple Choice:

- Given: $1/\log(\text{base } 2) a + 1/\log(\text{base } 3) a + 1/\log(\text{base } 4) a = 1$. Find a .
A. 10 C. 14
B. 18 D. 24
- If $x^2 + y^2 = 14xy$, then $\log[k(x + y)] = \frac{1}{2}(\log x + \log y)$ for some constant k . Find the value of k .
A. 0.50 C. 0.25
B. 0.30 D. 0.75
- If the roots of the quadratic equation $Ax^2 + Bx + C = 0$ are 3 and 2 and A , B and C are whole numbers, find the value of $A + B + C$.
A. 1 C. 2
B. -1 D. -2
- Find the value of x in the equation $2\log_a x - \log_a(x-1) - \log_a(x-2) = 0$.
A. $\frac{3}{4}$ C. $\frac{1}{2}$
B. $\frac{1}{3}$ D. $\frac{2}{3}$
- What is the least common multiple of 15 and 18?
A. 3 C. 5
B. 90 D. 270
- If $p - q = 5$ and $pq = \frac{k}{2}$, then $p^2 + q^2$ equals
A. k C. $25k$
B. $25 + k$ D. $\frac{k}{25}$
- Find k if $x^2 + 2(k + 2)x + 9k = 0$ has equal roots.
A. 2 C. 3
B. 0.5 D. 4
- Find the value of k in the quadratic equation $3x^2 - kx + x - 7k = 0$ if 3 is one of the roots.
A. 3 C. 18
B. 6 D. 12
- If $2x + 4y = 7x - 6y$, then $1/x : 1/y$ is
A. 2:1 C. 3:1
B. 1:2 D. 1:3
- Simplify: $[(xy)^n][(a - b)^{-1}][(a - b)^{(2n - 1)}]/[xy(a - b)^2]^{(n - 1)}$.
A. $a - b$ C. ab/xy
B. xy D. $ab(x - y)$
- A wife works three days then a day off while his husband works five days then a day off. If the couple has a day-off together today, how many days after will they have another day off together?
A. 15 days C. 14 days
B. 12 days D. 11 days
- Find the middle term of the expansion of $(x^2 - 5)^8$.
A. $43,750x^8$ C. $44,736x^8$
B. $45,852x^8$ D. $46,940x^8$
- If $x - 3$ is a factor of $kx^2 - 6x + 2kx - 12$, then what is the value of k ?
A. 2 C. 3
B. 1 D. -2
- The expression $x^4 + ax^3 + 5x^2 + bx + 6$ when divided by $x - 2$ leaves a remainder of 16 and when divided by $x + 1$ leaves a remainder of 10. Find a .
A. 7 C. -5
B. -7 D. 5
- Find the fourth term of $(a - 2x)^7$.
A. $120(a^4)(x^3)$
B. $-120(a^4)(x^3)$
C. $-280(a^4)(x^3)$
D. $280(a^4)(x^3)$
- Find the term involving x^9 in the expansion of $(x^2 + 2/x)^{12}$.
A. $24,339x^9$ C. $23,337x^9$
B. $25,344x^9$ D. $22,889x^9$
- Compute for x : $x = [\log(\text{base } c) a][\log(\text{base } a) b][\log(\text{base } b) c]$
A. 1 C. $a^{(b/c)}$
B. $b^{(c/a)}$ D. $c^{(a/b)}$
- Find the sum of the coefficients in the expansion $(x + 2y - z)^8$.
A. 128 C. 428
B. 256 D. 384
- Solve for x : $(\ln x)^2 - 2 \ln x^3 = -5$.
A. e and e^3 C. e and e^4
B. e and e^5 D. e and e^6
- If $x:y:z = 2:5:7$ and $4x - y + 2z = 51$, find z .
A. 21 C. 15
B. 6 D. 8
- Find the 6th term of $[1/2a - 3]^{16}$.
A. $-66939/256a^{11}$
B. $-33939/256a^{11}$
C. $-66339/128a^{11}$
D. $-66339/256a^{11}$
- Given: $\log_b y = 2x + \log_b x$. Which of the following is true?
A. $y = b^{2x}$ C. $y = 2xb$
B. $y = \frac{2x}{b}$ D. $y = xb^{2x}$
- Find the coefficient of $x(y^3)(z)$ in $(x + y + z)^5$.
A. 20 C. 25
B. 36 D. 30
- Find the sum of the exponents of the expansion $[(x^2) + 2/x]^{12}$.
A. 78 C. 23
B. 87 D. 22
- Evaluate $\{[(n + 1)!]^2/[n!(n - 1)!]\}$.
A. $n(n + 1)^2$ C. $n!$
B. $n(n + 1)$ D. $n + 1$